

Indian Standard DRINKING WATER - SPECIFICATION (First Revision) (Incorporating Amendment No. 1)

UDC 628.1.033

Bureau of Indian Standards Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002
Price Group 4 Copyright BIS 2003

Water Sectional Committee, CHD 13

FOREWORD

This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Water Sectional Committee had been approved by the Chemical Division Council.

This standard was originally published in 1983. The current revision has been undertaken to take into account the up to date information available about the nature and effect of various contaminants as also the new techniques for identifying and determining their concentration. In this revision, based on experience gained, additional requirements for alkalinity, aluminium and boron have been incorporated and the permissible limits for dissolved solids, nitrate and pesticides residues have been modified.

A report prepared by the World Health Organization in cooperation with the World Bank showed that in 1975, some 1230 million people were without safe water supplies. These appalling facts were central to the United Nations decision to declare an International Drinking Water Supply and Sanitation decade, beginning in 1981. Further, the VI Five-Year Plan of India had made a special provision for availability of safe drinking water for the masses. Therefore, this standard was prepared with the following objectives:

a) To assess the quality of water resources, and b) To check the effectiveness of water treatment and supply by the concerned authorities.

During VII Five-Year Plan, 55 mini mission districts were identified with a view to meet supply of water to all the problem villages. The VIII Five-Year Plan intends to provide safe drinking water to the rural mass. It also proposes to ensure supply of desired quality and required quantity of drinking water.

While preparing this standard, the Committee had taken note of the limited testing facilities available in the country. This standard, therefore, categorizes various characteristics as essential or desirable. The standard also mentions the desirable limit and indicates its background so that the implementing authorities may exercise their discretion, keeping in view the health of the people, adequacy of treatment, etc. All essential characteristics should be examined in routine. Besides, all desirable characteristics should be examined either when a doubt arises or the potability of water from a new source is to be established.

It has been recognised that often it is necessary to relax the specifications, especially when no alternate resources are available and therefore, to enable the experts to exercise their discretion, a maximum permissible limit has also been given.

In the case of virological examination, if not even one plaque-forming unit (PFU) of virus can be found in 1 litre of water, it can reasonably be assumed that the water is safe to drink. It would, however, be necessary to examine a sample of the order of 10 litres to obtain a proper estimation of the PFUs at this level. Such examinations cannot be made in ordinary control laboratories but there should be at least one laboratory in the country or region capable of carrying out virus examinations and also of pursuing further research on this subject.

The methods of test for various characteristics mentioned in this standard are currently under revision and their latest revision shall be used in testing.

In the formulation of this standard, assistance has been derived from the following publications:

a) International Standards for Drinking Water issued by World Health Organization, 1984, Geneva. b) Manual of Standards of Quality for Drinking Water Supplies, Indian Council of Medical Research, 1971, New Delhi. c) Manual on Water Supply and Treatment (third revision), Ministry of Urban Development, 1989, New Delhi.

This edition 2.1 incorporates Amendment No. 1 (January 1993). Side bar indicates modification of the text as the result of incorporation of the amendment.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

The standard prescribes the requirements for the essential and desirable characteristics required to be tested for ascertaining the suitability of water for drinking purpose.

2 REFERENCES

The Indian Standard listed in Annex A is a necessary adjunct to this standard.

3 CHARACTERISTICS

3.1 The test characteristics are given in Table 1 (see below, converted to a plain list).

3.2 Bacteriological Examination

3.2.1 Water in Distribution System

Ideally, all samples taken from the distribution system including consumers' premises should be free from coliform organisms. In practice, this is not always attainable, and the following standard for water collected in the distribution system is therefore recommended when tested in accordance with IS 1622 : 1981.

a) Throughout any year, 95 percent of samples should not contain any coliform organisms in 100 ml. b) No sample should contain *E. coli* in 100 ml. c) No sample should contain more than 10 coliform organisms per 100 ml. d) Coliform organisms should not be detectable in

100 ml of any two consecutive samples.

3.2.1.1 If any coliform organisms are found, the minimum action required is immediate resampling. The repeated finding of 1 to 10 coliform organisms in 100 ml, or the appearance of higher numbers in individual samples, suggests that undesirable material is gaining access to the water and measures should at once be taken to discover and remove the source of the pollution.

3.2.2 Unpipd Water Supplies

Where it is impracticable to supply water to consumers through a piped distribution network, and where untreated sources such as wells, boreholes and springs which may not be naturally pure have to be used, the requirements for piped supplies may not be attainable. In such circumstances, disinfection, although desirable, is not always practicable, and considerable reliance has to be placed on sanitary inspection and not exclusively on the results of bacteriological examination.

Everything possible should be done to prevent pollution of the water. Obvious sources of contamination should be removed from the immediate catchment area, special attention being given to the safe disposal of excrement. Wells and storage tanks should be protected by lining and covering; surface drainage should be diverted, erosion prevented and the surrounding area paved. Access of man and animals should be restricted by fencing, and should be so designed that fouling is prevented when drawing water. Although not supplied through pipes, water from such sources is likely to undergo further deterioration in quality during transport or storage before drinking. Containers used for water should be kept clean, covered and clear of the floor. The most important factor in achieving these objectives is to ensure the cooperation of the local community, and the importance of education in simple sanitary hygiene should be strongly stressed. In hospitals or medical clinics with such supplies, the value of some form of treatment is stressed.

3.2.2.1 Bacteriologically, the objective should be to reduce the coliform count to less than 10 per 100 ml, but more importantly, to ensure the absence of faecal coliform organisms. If these organisms are repeatedly found, or if sanitary inspection reveals obvious sources of pollution which cannot be avoided, then an alternative source of drinking water would be sought, whenever possible. Greater use should be made of protected ground-water sources and rain-water catchment which are more likely to meet requirements for potable water quality.

3.2.2.2 Although private sources of drinking water may be outside the jurisdiction of public health and water supply authorities, such supplies should still be of potable quality. The results of bacteriological tests and those of sanitary surveys should therefore be used to encourage improvement. Partial treatment may be necessary to remove turbidity even when coliform counts are low; and other quality criteria may dictate the need for treatment processes.

3.3 Virological Examination

3.3.1 It is theoretically possible that virus disease can be transmitted by water free from coliform organisms, but conclusive evidence that this has occurred is lacking.

3.3.2 None of the generally accepted sewage treatment methods yield virus-free effluent.

Although a number of investigators have found activated sludge treatment to be superior to trickling filters from this point of view, it seems possible that chemical precipitation methods will prove to be the most effective.

3.3.3 Virus can be isolated from raw water and from springs. Enterovirus, reovirus, and adenovirus have been found in water, the first named being the most resistant to chlorination. If enterovirus are absent from chlorinated water, it can be assumed that the water is safe to drink. Some uncertainty still remains about the virus of infectious hepatitis, since it has not so far been isolated, but in view of the morphology and resistance of enterovirus it is likely that, if they have been inactivated, hepatitis virus will have been inactivated also.

3.3.4 An exponential relationship exists between the rate of virus inactivation and the redox potential. A redox potential of 650 mV (measured between platinum and calomel electrodes) will cause almost instantaneous inactivation of even high concentrations of virus. Such a potential can be obtained with even a low concentration of free chlorine, but only with an extremely high concentration of combined chlorine. This oxidative inactivation may be achieved with a number of other oxidants also, for example, iodine, ozone, and potassium permanganate, but the effect of the oxidants will always be counteracted if reducing components, which are mainly organic, are present. As a consequence, the sensitivity of virus towards disinfectants will depend on the milieu just as much as on the particular disinfectant used.

3.3.5 Thus, in water in which free chlorine is present, active virus will generally be absent if coliform organisms are absent. In contrast, because the difference between the resistance of coliform organisms and of virus to disinfection by oxidants increases with increasing concentration of reducing components, for example organic matter, it cannot be assumed that the absence of viable coliform organisms implies freedom from active virus under circumstances where a free chlorine residual cannot be maintained. Sedimentation and slow sand filtration in themselves may contribute to the removal of virus from water.

3.3.6 In practice, 0.5 mg/l of free chlorine for one hour is sufficient to inactivate virus, even in water that was originally polluted.

3.4 Biological Examination

3.4.1 Biological examination is of value in determining the causes of objectionable tastes and odours in water and controlling remedial treatments, in helping to interpret the results of various chemical analyses, and in explaining the causes of clogging in distribution pipes and filters. In some instances, it may be of use in demonstrating that water from one source has been mixed with that from another.

3.4.2 The biological qualities of water are of greater importance when the supply has not undergone the conventional flocculation and filtration processes, since increased growth of methane-utilizing bacteria on biological slimes in pipes may then be expected, and the development of bryozoal growths such as *Plumatella* may cause operational difficulties.

3.4.3 Some of the animalcules found in water mains may be free-living in the water, but others such as *Dreissena* and *Asellus* are more or less firmly attached to the inside of the mains. Although these animalcules are not themselves pathogenic, they may harbour

pathogenic organisms or virus in their intestines, thus protecting these pathogens from destruction by chlorine.

3.4.4 Chlorination, at the dosages normally employed in waterworks, is ineffective against certain parasites, including amoebic cysts; they can be excluded only by effective filtration or by higher chlorine doses than can be tolerated without subsequent dechlorination.

Amoebiasis can be conveyed by water completely free from enteric bacteria; microscopic examination after concentration is, therefore, the only safe method of identification.

3.4.5 Strict precautions against back-syphonage and cross-connections are required if amoebic cysts are found in a distribution system containing tested water.

3.4.6 The cercariae of schistosomiasis can be detected by similar microscopic examination, but there is, in any case, no evidence to suggest that this disease is normally spread through piped water supplies.

3.4.7 The cyclops vector of the embryos of *Dracunculus medinensis*, which causes dracontiasis or Guinea-worm disease, can be found in open wells in a number of tropical areas. They are identifiable by microscopic examination. Such well supplies are frequently used untreated, but the parasite can be relatively easily excluded by simple physical improvements in the form of curbs, drainage, and apron surrounds, and other measures which prevent physical contact with the water source.

3.4.8 The drinking water shall be free from microscopic organisms such as algae, zooplanktons, flagellates, parasites and toxin-producing organisms. An illustrative (and not exhaustive) list is given in Annex B for guidance.

4 SAMPLING

Representative samples of water shall be drawn as prescribed in IS 1622 : 1981 and IS 3025 (Part 1) : 1987.

TABLE 1: TEST CHARACTERISTICS FOR DRINKING WATER (converted from table to a line-by-line list)

Each entry below lists, in order: Serial number; Substance or Characteristic; Requirement (Desirable Limit); Undesirable Effect Outside the Desirable Limit; Permissible Limit in the Absence of an Alternate Source; Method of Test (IS reference); Remarks.

ESSENTIAL CHARACTERISTICS

i) Colour, Hazen units, Max

- Desirable limit: 5
- Undesirable effect: Above 5, consumer acceptance decreases
- Permissible limit (no alternate source): 25
- Test method: IS 3025 (Part 4) : 1983
- Remarks: Extended to 25 only if toxic substances are not suspected, in absence of alternate sources

ii) Odour

- Desirable limit: Unobjectionable
- Undesirable effect: Not specified
- Permissible limit: Not specified
- Test method: IS 3025 (Part 5) : 1983
- Remarks: (a) Test cold and when heated; (b) Test at several dilutions

iii) Taste

- Desirable limit: Agreeable
- Undesirable effect: Not specified
- Permissible limit: Not specified
- Test method: IS 3025 (Part 7 and 8) : 1984
- Remarks: Test to be conducted only after safety has been established

iv) Turbidity, NTU, Max

- Desirable limit: 5
- Undesirable effect: Above 5, consumer acceptance decreases
- Permissible limit: 10
- Test method: IS 3025 (Part 10) : 1984
- Remarks: None

v) pH value

- Desirable limit: 6.5 to 8.5
- Undesirable effect: Beyond this range the water will affect the mucous membrane and/or water supply system
- Permissible limit: No relaxation
- Test method: IS 3025 (Part 11) : 1984
- Remarks: None

vi) Total hardness (as CaCO₃), mg/l, Max

- Desirable limit: 300
- Undesirable effect: Encrustation in water supply structure and adverse effects on domestic use
- Permissible limit: 600
- Test method: IS 3025 (Part 21) : 1983
- Remarks: None

vii) Iron (as Fe), mg/l, Max

- Desirable limit: 0.3
- Undesirable effect: Beyond this limit taste/appearance are affected, has adverse effect on domestic uses and water supply structures, and promotes iron bacteria
- Permissible limit: 1.0
- Test method: Clause 32 of IS 3025 : 1964

- Remarks: None

viii) Chloride (as Cl), mg/l, Max

- Desirable limit: 250
- Undesirable effect: Beyond this limit, taste, corrosion and palatability are affected
- Permissible limit: 1000
- Test method: IS 3025 (Part 32) : 1988
- Remarks: None

ix) Residual, free chlorine, mg/l, Min

- Desirable limit: 0.2
- Undesirable effect: Not specified
- Permissible limit: Not specified
- Test method: IS 3025 (Part 26) : 1986
- Remarks: To be applicable only when water is chlorinated. Tested at consumer end. When protection against viral infection is required, it should be minimum 0.5 mg/l.

x) Fluoride (as F), mg/l, Max

- Desirable limit: 1.0
- Undesirable effect: Fluoride may be kept as low as possible. High fluoride may cause fluorosis
- Permissible limit: 1.5
- Test method: Clause 23 of IS 3025 : 1964
- Remarks: None

DESIRABLE CHARACTERISTICS

xi) Dissolved solids, mg/l, Max

- Desirable limit: 500
- Undesirable effect: Beyond this palatability decreases and may cause gastro-intestinal irritation
- Permissible limit: 2000
- Test method: IS 3025 (Part 16) : 1984
- Remarks: None

xii) Calcium (as Ca), mg/l, Max

- Desirable limit: 75
- Undesirable effect: Encrustation in water supply structure and adverse effects on domestic use
- Permissible limit: 200
- Test method: IS 3025 (Part 40) : 1991
- Remarks: None

xiii) Magnesium (as Mg), mg/l, Max

- Desirable limit: 30
- Undesirable effect: Encrustation of water supply structure and adverse effects on domestic use
- Permissible limit: 100
- Test method: Clauses 16, 33, 34 of IS 3025 : 1964
- Remarks: None

xiv) Copper (as Cu), mg/l, Max

- Desirable limit: 0.05
- Undesirable effect: Astringent taste, discoloration and corrosion of pipes, fittings and utensils will be caused beyond this
- Permissible limit: 1.5
- Test method: Clause 36 of IS 3025 : 1964
- Remarks: None

xv) Manganese (as Mn), mg/l, Max

- Desirable limit: 0.1
- Undesirable effect: Beyond this limit taste/appearance are affected, has adverse effect on domestic uses and water supply structures
- Permissible limit: 0.3
- Test method: Clause 35 of IS 3025 : 1964
- Remarks: None

xvi) Sulphate (as SO₄), mg/l, Max

- Desirable limit: 200
- Undesirable effect: Beyond this causes gastro-intestinal irritation when magnesium or sodium are present
- Permissible limit: 400 (see remarks)
- Test method: IS 3025 (Part 24) : 1986
- Remarks: May be extended up to 400 provided magnesium (as Mg) does not exceed 30

xvii) Nitrate (as NO₃), mg/l, Max

- Desirable limit: 45
- Undesirable effect: Beyond this, methaemoglobinemia takes place
- Permissible limit: 100
- Test method: IS 3025 (Part 34) : 1988
- Remarks: None

xviii) Phenolic compounds (as C₆H₅OH), mg/l, Max

- Desirable limit: 0.001
- Undesirable effect: Beyond this, it may cause objectionable taste and odour

- Permissible limit: 0.002
- Test method: Clause 54 of IS 3025 : 1964
- Remarks: None

xix) Mercury (as Hg), mg/l, Max

- Desirable limit: 0.001
- Undesirable effect: Beyond this, the water becomes toxic
- Permissible limit: No relaxation (see note below the table)
- Test method: Mercury ion analyser
- Remarks: To be tested when pollution is suspected

xx) Cadmium (as Cd), mg/l, Max

- Desirable limit: 0.01
- Undesirable effect: Beyond this, the water becomes toxic
- Permissible limit: No relaxation (see note below the table)
- Test method: Not specified
- Remarks: To be tested when pollution is suspected

xxi) Selenium (as Se), mg/l, Max

- Desirable limit: 0.01
- Undesirable effect: Beyond this, the water becomes toxic
- Permissible limit: No relaxation
- Test method: Clause 28 of IS 3025 : 1964
- Remarks: To be tested when pollution is suspected

xxii) Arsenic (as As), mg/l, Max

- Desirable limit: 0.05
- Undesirable effect: Beyond this, the water becomes toxic
- Permissible limit: No relaxation
- Test method: IS 3025 (Part 37) : 1988
- Remarks: To be tested when pollution is suspected

xxiii) Cyanide (as CN), mg/l, Max

- Desirable limit: 0.05
- Undesirable effect: Beyond this limit, the water becomes toxic
- Permissible limit: No relaxation
- Test method: IS 3025 (Part 27) : 1986
- Remarks: To be tested when pollution is suspected

xxiv) Lead (as Pb), mg/l, Max

- Desirable limit: 0.05
- Undesirable effect: Beyond this limit, the water becomes toxic

- Permissible limit: No relaxation (see note below the table)
- Test method: Not specified
- Remarks: To be tested when pollution or plumbosolvency is suspected

xxv) Zinc (as Zn), mg/l, Max

- Desirable limit: 5
- Undesirable effect: Beyond this limit it can cause astringent taste and an opalescence in water
- Permissible limit: 15
- Test method: Clause 39 of IS 3025 : 1964
- Remarks: To be tested when pollution is suspected

xxvi) Anionic detergents (as MBAS), mg/l, Max

- Desirable limit: 0.2
- Undesirable effect: Beyond this limit it can cause a light froth in water
- Permissible limit: 1.0
- Test method: Methylene-blue extraction method
- Remarks: To be tested when pollution is suspected

xxvii) Chromium (as Cr⁶⁺), mg/l, Max

- Desirable limit: 0.05
- Undesirable effect: May be carcinogenic above this limit
- Permissible limit: No relaxation
- Test method: Clause 38 of IS 3025 : 1964
- Remarks: To be tested when pollution is suspected

xxviii) Polynuclear aromatic hydrocarbons (as PAH), micrograms/l, Max

- Desirable limit: Not specified (dash in source table)
- Undesirable effect: May be carcinogenic
- Permissible limit: Not specified
- Test method: Not specified
- Remarks: None

xxix) Mineral oil, mg/l, Max

- Desirable limit: 0.01
- Undesirable effect: Beyond this limit undesirable taste and odour after chlorination take place
- Permissible limit: 0.03
- Test method: Gas chromatographic method
- Remarks: To be tested when pollution is suspected

xxx) Pesticides, mg/l, Max

- Desirable limit: Absent
- Undesirable effect: Toxic
- Permissible limit: 0.001
- Test method: Not specified
- Remarks: None

xxxix) Radioactive materials: a) Alpha emitters, Bq/l, Max - Desirable limit: Not specified - Permissible limit: 0.1 - Test method: Clause 58 of IS 3025 : 1964 b) Beta emitters, pCi/l, Max - Desirable limit: Not specified - Permissible limit: 1 - Test method: Not specified

xxxii) Alkalinity, mg/l, Max

- Desirable limit: 200
- Undesirable effect: Beyond this limit taste becomes unpleasant
- Permissible limit: 600
- Test method: Clause 13 of IS 3025 : 1964
- Remarks: None

xxxiii) Aluminium (as Al), mg/l, Max

- Desirable limit: 0.03
- Undesirable effect: Cumulative effect is reported to cause dementia
- Permissible limit: 0.2
- Test method: Clause 31 of IS 3025 : 1964
- Remarks: None

xxxiv) Boron, mg/l, Max

- Desirable limit: 1
- Undesirable effect: Not specified
- Permissible limit: 5
- Test method: Clause 29 of IS 3025 : 1964
- Remarks: None

NOTE (from below the table): For items xix, xx and xxiv (mercury, cadmium, and lead), the atomic absorption spectrophotometric method may be used.

ANNEX A (Clause 2) - LIST OF REFERRED INDIAN STANDARDS

- IS 1622 : 1981 - Methods of sampling and microbiological examination of water (first revision)
- IS 3025 : 1964 - Methods of sampling and test (physical and chemical) for water used in industry
- IS 3025 (Part 1) : 1987 - Methods of sampling and test (physical and chemical) for water and waste water: Part 1 Sampling (first revision)
- IS 3025 (Part 4) : 1983 - Part 4 Colour (first revision)
- IS 3025 (Part 5) : 1983 - Part 5 Odour (first revision)

- IS 3025 (Part 7) : 1984 - Part 7 Taste threshold (first revision)
 - IS 3025 (Part 8) : 1984 - Part 8 Taste rating (first revision)
 - IS 3025 (Part 10) : 1984 - Part 10 Turbidity (first revision)
 - IS 3025 (Part 11) : 1983 - Part 11 pH value (first revision)
 - IS 3025 (Part 16) : 1984 - Part 16 Filterable residue (total dissolved solids) (first revision)
 - IS 3025 (Part 21) : 1983 - Part 21 Total hardness (first revision)
 - IS 3025 (Part 24) : 1986 - Part 24 Sulphates (first revision)
 - IS 3025 (Part 26) : 1986 - Part 26 Chlorine residual (first revision)
 - IS 3025 (Part 27) : 1986 - Part 27 Cyanide (first revision)
 - IS 3025 (Part 32) : 1988 - Part 32 Chloride (first revision)
 - IS 3025 (Part 34) : 1988 - Part 34 Nitrogen (first revision)
 - IS 3025 (Part 37) : 1988 - Part 37 Arsenic (first revision)
-

ANNEX B (Clause 3.4.8) - ILLUSTRATIVE LIST OF MICROSCOPIC ORGANISMS WHICH MAY BE PRESENT IN WATER

Each entry below lists: Classification group; Organism names; Habitat; Effect and significance.

1. ALGAE

a) Chlorophyceae

- Species of Coelastrum, Gomphosphaerium, Micractinium, Mougeotia, Oocystis, Euastrum, Scenedesmus, Actinastrum, Gonium, Eudorina, Pandorina, Pediasstrum, Zygnema, Chlamydomonas, Carteria, Chlorella, Chroococcus, Spirogyra, Tetraedron, Chlorogonium, Stigeoclonium
 - Habitat: Polluted water, impounded sources
 - Effect: Impart colouration
- Species of Pandorina, Volvox, Gomphosphaerium, Staurastrum, Hydrodictyon, Nitella
 - Habitat: Polluted waters
 - Effect: Produce taste and odour
- Species of Rhizoclonium, Cladotrix, Ankistrodesmus, Ulothrix, Micrasterias, Chromulina
 - Habitat: Clean water
 - Effect: Indicate clean condition
- Species of Chlorella, Tribonema, Closterium, Spirogyra, Palmella
 - Habitat: Polluted waters, impounded sources
 - Effect: Clog filters and create operational difficulties

b) Cyanophyceae

- Species of Anacystis and Cylindrospermum

- Habitat: Polluted waters
- Effect: Cause water bloom and impart colour
- Species of Anabaena, Phormidium, Lyngbya, Arthrospira, Oscillatoria
 - Habitat: Polluted waters
 - Effect: Impart colour
- Species of Anabaena, Anacystis, Aphanizomenon
 - Habitat: Polluted waters, impounded sources
 - Effect: Produce taste and odour
- Species of Anacystis, Anabaena, Coelosphaerium, Gloeotrichia, Aphanizomenon
 - Habitat: Polluted waters
 - Effect: Toxin producing
- Species of Anacystis, Rivularia, Oscillatoria, Anabaena
 - Habitat: Polluted waters
 - Effect: Clog filters
- Species of Rivularia
 - Habitat: Calcareous waters and also rocks
 - Effect: Bores rocks and calcareous strata and causes matted growth
- Species of Agmenellum, Microcoleus, Lemanea
 - Habitat: Clean waters
 - Effect: Indicators of purification

c) Diatoms (Bacillariophyceae)

- Species of Fragillaria, Stephanodiscus, Stauroneis
 - Effect: Cause discoloration
- Species of Asterionella, Tabellaria
 - Habitat: Hill streams, high altitude, torrential and temperate waters
 - Effect: Taste and odour producing, clog filters
- Species of Synedra and Fragillaria
 - Habitat: Polluted waters
 - Effect: Taste and odour producing
- Species of Nitzschia, Gomphonema
 - Habitat: Moderately polluted waters
 - Effect: Cause discoloration
- Species of Cymbella, Synedra, Melosira, Navicula, Cyclotella, Fragillaria, Diatoma, Pleurosigma
 - Habitat: Rivers and streams, impounded sources
 - Effect: Clog filters and cause operational difficulties
- Species of Pinnularia, Surirella, Cyclotella, Meridion, Cocconeis

- Habitat: Clean waters
- Effect: Indicators of purification

d) Xanthophyceae

- Species of Botryococcus
 - Habitat: Hill streams, high altitudes and temperate waters
 - Effect: Produces coloration

2. ZOOPLANKTON

a) Protozoa

- Amoeba, Giardia lamblia, Arcella, Diffugia, Actinophrys
 - Habitat: Polluted waters
 - Effect: Pollution indicators
- Endamoeba histolytica
 - Habitat: Sewage and activated sludge
 - Effect: Parasitic and pathogenic

b) Ciliates

- Paramecium, Vorticella, Carchesium, Stentor, Colpidium, Coleps, Euplotes, Colpoda, Bodo
 - Habitat: Highly polluted waters, sewage and activated sludge
 - Effect: Bacteria eaters

c) Crustacea

- Bosmina, Daphnia
 - Habitat: Stagnant polluted waters
 - Effect: Indicators of pollution
- Cyclops
 - Habitat: Step wells in tropical climate
 - Effect: Carrier host of guinea worm

3. ROTIFERS

a) Rotifers

- Anurea, Rotaria, Philodina
 - Habitat: Polluted and algae-laden waters
 - Effect: Feed on algae

b) Flagellates

- Ceratium, Glenodinium, Peridinium, Dinobryon
 - Habitat: Rocky strata, iron-bearing and acidic waters

- Effect: Impart colour and fishy taste
- Euglena, Phacus
 - Habitat: Polluted waters
 - Effect: Impart colour

c) Miscellaneous Organisms

- Sponges, Hydra
 - Habitat: Fresh water
 - Effect: Clog filters and affect purification systems
- Tubifex, Eristalis, Chironomids
 - Habitat: Highly polluted waters, sewage and activated sludge, and bottom deposits
 - Effect: Clog filters and render water unaesthetic
- Plumatella
 - Habitat: Polluted waters
 - Effect: Produces biological slimes and causes filter operational difficulties
- Dreissena, Asellus
 - Habitat: Polluted waters
 - Effect: Harbour pathogenic organisms

ABOUT BUREAU OF INDIAN STANDARDS (BIS)

BIS is a statutory institution established under the Bureau of Indian Standards Act, 1986 to promote harmonious development of the activities of standardization, marking and quality certification of goods and attending to connected matters in the country.

Copyright: BIS holds the copyright of all its publications. No part of these publications may be reproduced in any form without prior permission in writing of BIS. This does not preclude the free use, in the course of implementing the standard, of necessary details such as symbols and sizes, type or grade designations. Enquiries relating to copyright should be addressed to the Director (Publications), BIS.

Review of Indian Standards: Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of the BIS Catalogue and Standards: Monthly Additions.

This Indian Standard has been developed from Doc No. ET 13 (0046).

Amendments Issued Since Publication: Amendment No. 1, issued January 1993.

Bureau of Indian Standards Headquarters: Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002. Telephones: 323 01 31, 323 33 75, 323 94 02 Telegrams: Manaksanstha (common to all offices)

Regional Offices:

- Central: Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002. Telephones: 323 76 17, 323 38 41
- Eastern: 1/14 C.I.T. Scheme VII M, V.I.P. Road, Kankurgachi, Kolkata 700054. Telephones: 337 84 99, 337 85 61, 337 86 26, 337 91 20
- Northern: SCO 335-336, Sector 34-A, Chandigarh 160022. Telephones: 60 38 43, 60 20 25
- Southern: C.I.T. Campus, IV Cross Road, Chennai 600113. Telephones: 235 02 16, 235 04 42, 235 15 19, 235 23 15
- Western: Manakalaya, E9 MIDC, Marol, Andheri (East), Mumbai 400093. Telephones: 832 92 95, 832 78 58, 832 78 91, 832 78 92

Branches: Ahmedabad, Bangalore, Bhopal, Bhubaneshwar, Coimbatore, Faridabad, Ghaziabad, Guwahati, Hyderabad, Jaipur, Kanpur, Lucknow, Nagpur, Nalagarh, Patna, Pune, Rajkot, Thiruvananthapuram, Vishakhapatnam.